

Amazon Product Co-Purchasing Network Analysis

Introduction

This project analyzes an Amazon product co-purchasing network using Gephi. The dataset was imported as an undirected graph, where nodes represent products and edges represent co-purchasing relationships between products. Network analysis techniques such as degree analysis, PageRank, betweenness centrality, and community detection were applied to identify influential products and understand the structure of the network.

Methodology

Dataset Import

The CSV dataset was imported into Gephi as an undirected graph using the Import Spreadsheet option. The graph contains:

- Nodes: 1000
- Edges: 2178

Graph Visualization

The Yifan Hu layout algorithm was applied to organize the network structure for better readability. Node sizes were adjusted according to degree values, while edge thickness was reduced to improve visualization clarity.

Network Metrics

The following network analysis algorithms were applied:

- Degree Analysis
- Betweenness Centrality
- PageRank
- Modularity (Community Detection)

Community Detection

The Modularity algorithm was used to detect communities within the network. Nodes were colored according to their modularity class to visually identify product clusters.

Degree Analysis

Degree measures the number of connections a node has with other nodes in the network.

Top 5 Nodes by Degree

Node ID	Degree
458358	324
38365	151
175102	92
498702	46
37981	46

Analysis

Node 458358 is the dominant hub in the network with a degree of 324, indicating that it is connected to a very large number of products. High-degree nodes act as central hubs and represent highly co-purchased products within the Amazon network.

Betweenness Centrality Analysis

Betweenness centrality identifies nodes that act as bridges connecting different communities in the network.

Top 5 Nodes by Betweenness Centrality

Node ID	Betweenness Centrality
274481	2991.0

306687	2988.0
521666	2970.009302
156600	2945.864394
462447	2919.688789

Analysis

Nodes with high betweenness centrality serve as connectors between separate product communities. These nodes are important because they facilitate interactions and maintain connectivity across the network.

PageRank Analysis

PageRank identifies globally influential nodes based on the importance of their neighboring connections.

Top 5 Nodes by PageRank

Node ID	PageRank
123573	0.001822
475968	0.001815
524987	0.001789
109245	0.001784
321592	0.001754

Analysis

The PageRank results indicate the most influential products in the network. These products are connected to other highly connected products, making them globally important within the co-purchasing structure.

Community Detection

The Modularity algorithm identified several product communities within the Amazon co-purchasing network. Different colors were assigned to each modularity class to visually distinguish the communities.

Analysis

The detected communities represent groups of products that are frequently purchased together. Community structures reveal hidden relationships and purchasing patterns among Amazon products.

Filtering and Highlighting

Filtering techniques were applied using the Degree Range filter to isolate high-degree nodes. The top 5 degree nodes were manually highlighted by:

- Increasing node size
- Coloring nodes black

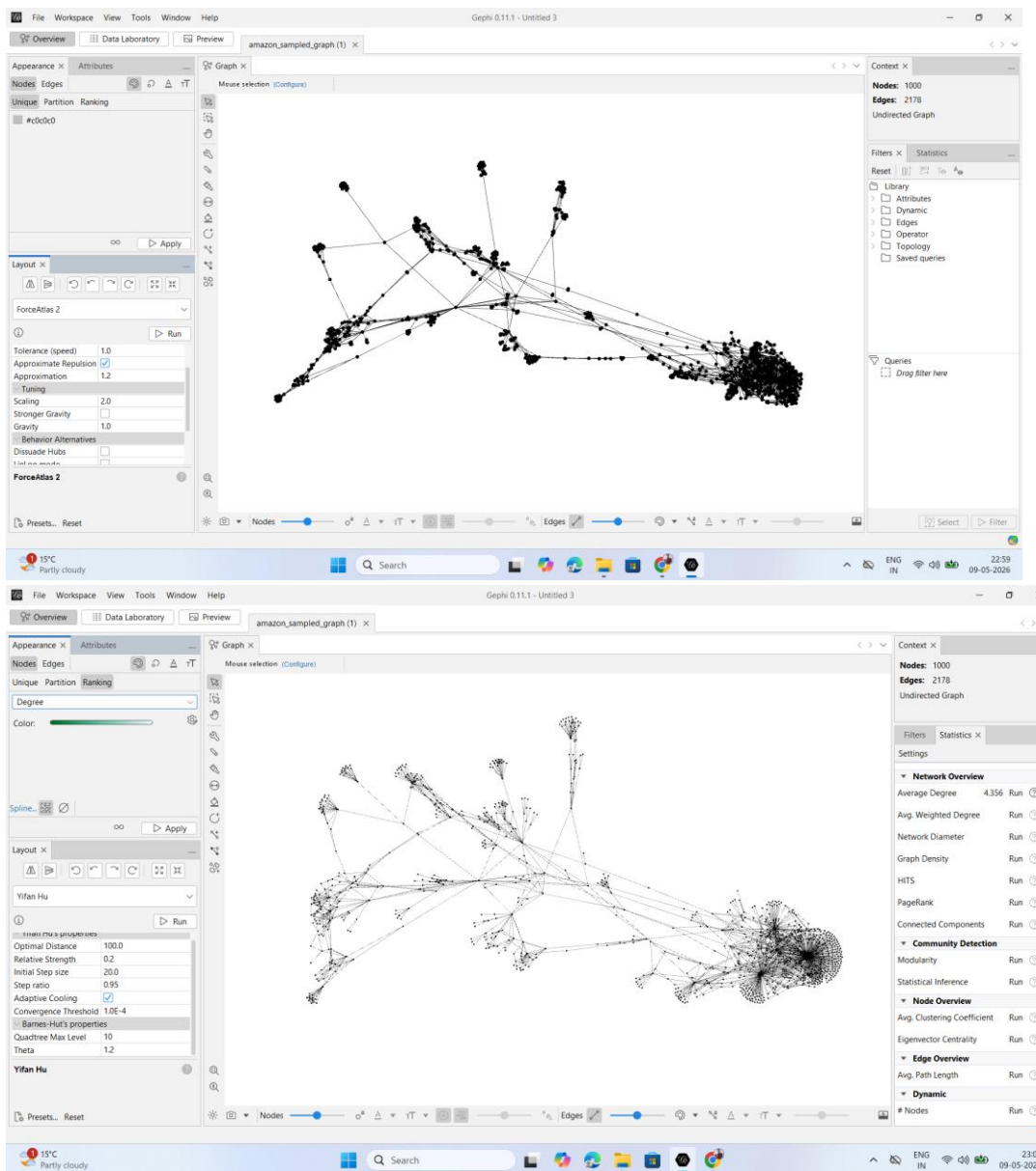
This helped emphasize the most influential hub products within the network.

Conclusion

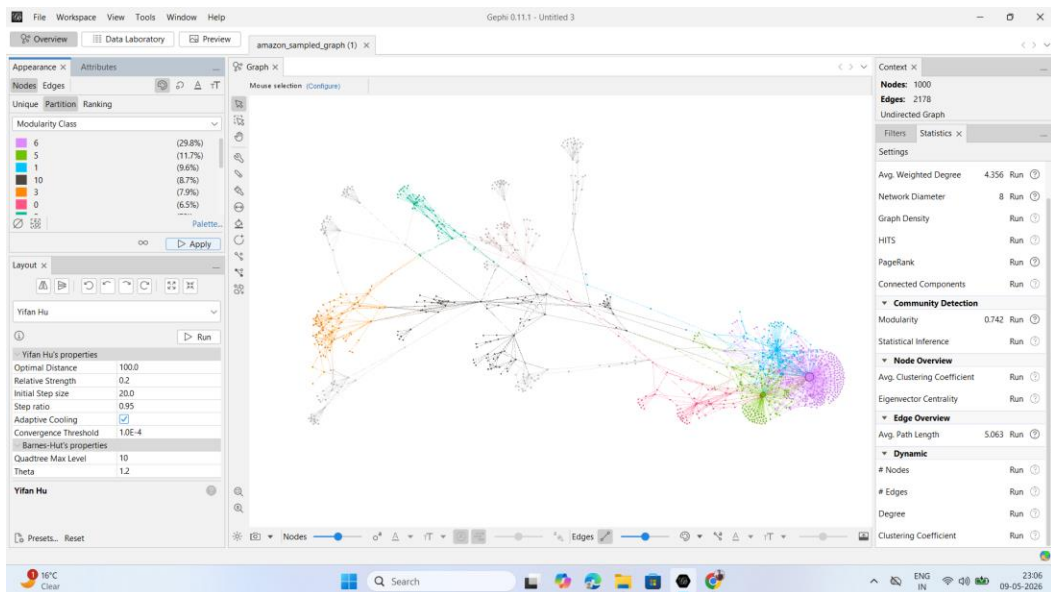
The Amazon co-purchasing network demonstrates a community-based structure with several highly influential hub nodes. Degree analysis identified products with the highest connectivity, while betweenness centrality highlighted bridge nodes connecting separate communities. PageRank revealed globally influential products within the network.

Overall, the analysis shows that a small number of products dominate the network structure and significantly influence co-purchasing behavior across the Amazon product ecosystem.

1. Original Network Visualization



2. Community Detection Visualization



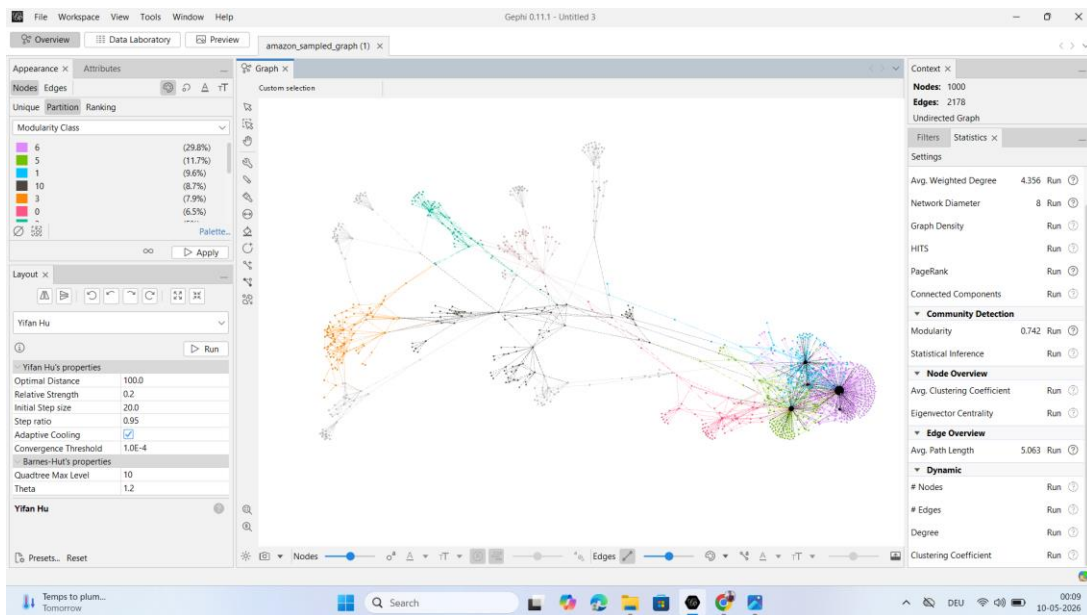
3. Betweenness Centrality Table

Nodes	Edges	Configuration	Add node	Add edge	Search/Replace	Import Spreadsheet	Export table	More actions	Filter:	Id
Id	Label	Interval	Degree	PageRank	Eccentricity	Closeness Centrality	Harmonic Closeness Centrality	Betweenness Centrality	Weighted Degree	Modularity Class
274481		4	0.001576	6.0	0.191233	0.200951	2991.0	4.0	10	
306687		5	0.001183	5.0	0.222991	0.234718	2988.0	10	10	
521666		6	0.001416	6.0	0.187114	0.202536	2970.009302	6.0	8	
156600		5	0.001113	8.0	0.230769	0.290128	2945.864394	5.0	0	
462447		8	0.001753	7.0	0.268621	0.337566	2919.688789	8.0	1	
385537		5	0.001117	6.0	0.186068	0.198632	2901.235714	5.0	10	
509665		10	0.001951	7.0	0.184046	0.202817	2851.255115	10.0	2	
211669		5	0.001436	7.0	0.24348	0.277506	2559.165079	5.0	6	

4. Highlighted Top Degree Nodes

Id	Label	Interval	Degree	
458358			324	0.06
38365			151	0.03
175102			92	0.01
498702			46	0.00
37981			46	0.00
521221			41	0.00
222939			32	0.00
306602			27	0.00

5. Filtered High-Degree Node Visualization



6. Final Network Layout in Preview Mode

